

Phishing Email Analysis Report

Objective

The objective of this task was to analyze a suspected phishing email and identify characteristics commonly associated with phishing attacks. Various email security checks and domain reputation tests were performed using MXToolbox.

1. Sample Phishing Email

Sender Email Address

security-update@paypaI-support.com

Subject

Urgent: Your Account Will Be Suspended!

Email Body

The email claims that unusual activity was detected on the recipient's account and requests immediate verification through a provided link. The message warns that the account will be suspended within 24 hours if action is not taken.

2. Sender Address Analysis

The sender address attempts to impersonate PayPal by using a deceptive domain name:

security-update@paypaI-support.com

Observations:

- The domain resembles PayPal's legitimate domain.
- The domain is not owned by PayPal.
- The sender uses a look-alike domain to deceive recipients.

Result

The sender address appears spoofed and is considered suspicious.

3. Email Header and Domain Security Analysis

MXToolbox was used to analyze the domain configuration and email security records.

Findings

DMARC Check

- No DMARC Record Found

DMARC helps prevent email spoofing and phishing attacks. The absence of a DMARC record indicates weak email authentication.

SPF Check

- No SPF Record Found

SPF verifies authorized mail servers. Missing SPF records increase the risk of email spoofing.

SMTP Server Test

- SMTP Connection Failed
- Unable to connect after 15 seconds

This indicates poor or suspicious mail server configuration.

Result

The domain lacks proper email authentication and security controls.

4. Suspicious Links Analysis

Link Found

<http://paypal-security-check.com/login>

Observations

- The URL does not belong to PayPal.
- Uses HTTP instead of HTTPS.
- Designed to mimic a legitimate PayPal login page.

Result

The link is highly suspicious and may be used to steal user credentials.

5. Urgent or Threatening Language

The email contains several urgency-based statements:

- "Your account will be suspended within 24 hours."
- "Verify your information immediately."
- "Failure to act may result in permanent account closure."

Result

These statements are intended to create fear and pressure the recipient into acting quickly without verifying authenticity.

6. URL Mismatch Analysis

The email claims to represent PayPal but directs users to:

paypal-security-check.com

instead of the official PayPal website:

paypal.com

Result

The mismatch between the claimed organization and the actual destination URL is a strong indicator of phishing.

7. Grammar and Writing Analysis

Observations

- Uses generic greeting: "Dear Customer"
- No personalization
- Unprofessional and suspicious tone

Result

Although major spelling mistakes are absent, the generic wording and lack of personalization are common phishing indicators.

8. Domain Reputation Analysis

MXToolbox Blacklist Check Results:

- Domain listed on Spamhaus DBL blacklist
- Domain reputation marked as suspicious

Result

The domain has been identified by security services as potentially malicious.

Summary of Findings

Phishing Indicator	Status
Spoofed Sender Address	Detected
Suspicious Domain	Detected
Missing SPF Record	Detected
Missing DMARC Record	Detected
SMTP Failure	Detected
Blacklisted Domain	Detected
Suspicious URL	Detected
URL Mismatch	Detected
Urgent Language	Detected
Generic Greeting	Detected

Conclusion

Based on the analysis, the email exhibits multiple characteristics of a phishing attack. The sender uses a deceptive domain name, directs users to a suspicious website, employs urgent language, lacks proper email authentication records, and is associated with a blacklisted domain.

The collected evidence strongly suggests that this email is a phishing attempt intended to trick recipients into disclosing sensitive information such as login credentials or financial details.

Final Verdict

Classification: PHISHING EMAIL

All required analysis tasks were successfully completed, and sufficient evidence was gathered to support the conclusion.